

Challenge (Habanero)

- Q1.** A building has 24 floors. If 3 elevators can carry people to every 4th floor, how many stops will each elevator make?
- (A) 4
(B) 6
(C) 8
(D) 12
(E) 16
- Q2.** A recipe for making cookies requires 15 cups of flour. If you want to make 3 batches of cookies, how many cups of flour will you need in total, if 3 cups can be bought in 1 package?
- (A) 5
(B) 10
(C) 12
(D) 15
(E) 45
- Q3.** If a number is divisible by 2 and 9, it must also be divisible by...
- (A) 3
(B) 4
(C) 5
(D) 6
(E) 18
- Q4.** What is the remainder when 43 is divided by 5?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) 3
- Q5.** In a bookshelf with 36 books, if the books are to be packed in boxes that hold 4 books each, how many boxes are needed?
- (A) 8
(B) 9
(C) 10
(D) 12
(E) 6
- Q6.** The sum of the digits of a number is 9. If the number is divisible by 3, it must also be divisible by...
- (A) 9
(B) 6
(C) 5
(D) 10
(E) 8
- Q7.** A floor is 12 meters long. If a carpet that is 3 meters long is laid along its length, what fraction of the floor will be covered?
- (A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{1}{4}$
(D) $\frac{3}{4}$
(E) $\frac{1}{6}$
- Q8.** What is the result of $27 \div 3$?
- (A) 8 remainder 3
(B) 9
(C) 10
(D) 12
(E) 6 remainder 1

Open Ended Questions (FRQ)

- Q9.** In a construction project, 48 bags of cement are required. If the bags are to be transported in trucks that can carry 8 bags each, how many trucks are needed? Show your calculation and explain your reasoning.
- Q10.** A chef needs to package 50 cookies into boxes that hold 5 cookies each. How many boxes are required? If the chef has 3 large boxes that can hold 15 cookies each, how many cookies will be left over after filling the large boxes? Show your calculation and explain your reasoning.

Chef's Tips

- Chef's Tips: Read each question carefully
- Chef's Tips: Check your calculations
- Chef's Tips: Use the back of the page for scratch work if needed